Skin Lesion Segmentation in Dermoscopic Images

A comparison of three classical computer-vision methods

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Revised November 1, 2026

Abstract

Melanoma is the most aggressive form of skin cancer, and its prognosis depends closely on how early it is detected. Automated delineation of the lesion is the first step of any computer-aided diagnosis system, since every downstream clinical measurement (asymmetry, border irregularity, colour) is computed from that border. This report presents a segmentation pipeline built *without* deep learning, under the explicit constraint of using classical image-processing techniques only. After a preprocessing stage removing the dermoscope frame and body hair, three segmentation methods are compared on 20 annotated dermoscopic images: LBP Clustering and Statistical Region Merging, reimplemented from their source publications, and a multi-channel Otsu thresholding of our own design. LBP Clustering gives the best raw masks, and a shape-aware convex hull raises every method further. We report where each method fails and why.

1 Introduction

Melanoma is a major public health concern: in the United States alone, about 105,000 invasive melanomas are expected to be diagnosed in 2025 and about 8,400 people to die of the disease [7]. Survival depends strongly on how quickly the lesion is identified and treated, so any technological improvement in screening tools can reduce mortality.

Dermoscopy is the reference imaging technique. By eliminating surface reflections it reveals subsurface structures, improving diagnostic sensitivity over a simple clinical examination. Interpreting these images nevertheless remains difficult and subject to inter-operator variability, even among experienced dermatologists. Computer-aided diagnosis (CAD) systems therefore matter: they offer a more objective, standardised and reproducible assessment. Such systems rest on segmentation algorithms whose role is to delineate the pathological area precisely.

This project automates that critical isolation step for pigmented lesions in dermoscopic images, under one specific constraint: classical image-processing techniques only, with no deep learning. The constraint is deliberate. A segmentation network needs annotated training data that a twenty-image study does not have, and its failures are hard to attribute; a classical pipeline runs without training and each of its stages can be inspected in isolation, which is what allows the failure analysis in Section 4.

The report first details the preprocessing needed to clean the images of their artefacts, then presents the implementation and comparison of three segmentation methods, and finally evaluates them against expert ground truth.

2 Preprocessing of dermoscopic images

2.1 Frame removal

Many dermoscopic images show a dark frame around the useful area, caused by the acquisition device or by a physical mask (vignetting). These black borders can be mistaken for the lesion, or disturb the segmentation algorithms by creating artificial contrast. Removing the frame before segmentation is therefore essential.

2.1.1 First approach: rectangular scanning

We initially implemented an iterative cropping algorithm scanning rows and columns from the image borders inwards. The algorithm computed the proportion of black pixels on each line; as soon as that proportion fell below a threshold (indicating skin), scanning stopped and defined the crop limits.

Observed limitations. This method implicitly assumed the region of interest was rectangular and aligned with the image borders. On several images of our dataset, however, dermoscopes produce a circular field of view. Dark corners survived the crop, which led us to a more robust approach accounting for the circular geometry.

2.1.2 Second approach: morphological segmentation and circular masking

Instead of removing borders line by line, the geometric approach identifies the central skin area as an *object* and masks out everything else. The algorithm assumes the illuminated skin forms a bright disc near the image centre, and proceeds as follows:

1. **Border analysis.** A preliminary inspection of a band covering 5 % of the image's shorter side gives the proportion of black pixels. If that proportion is low the image is considered frameless and left untouched. That matters, since cropping a frameless image would only destroy data.

2. **Global thresholding and connected components.** The image is converted to grayscale and thresholded to separate the bright skin from the dark background. Connected-component analysis retains the largest region, discarding noise and isolated reflections.

3. **Geometric modelling.** The main component is approximated by a disc whose centroid and radius are computed. The radius is shrunk by 10 % to stay clear of the blurred rim.

4. **Masking and cropping.** Everything outside the disc is replaced by white (255) rather than black, so the discarded area cannot later be mistaken for a dark lesion. A rectangular crop around the disc follows.

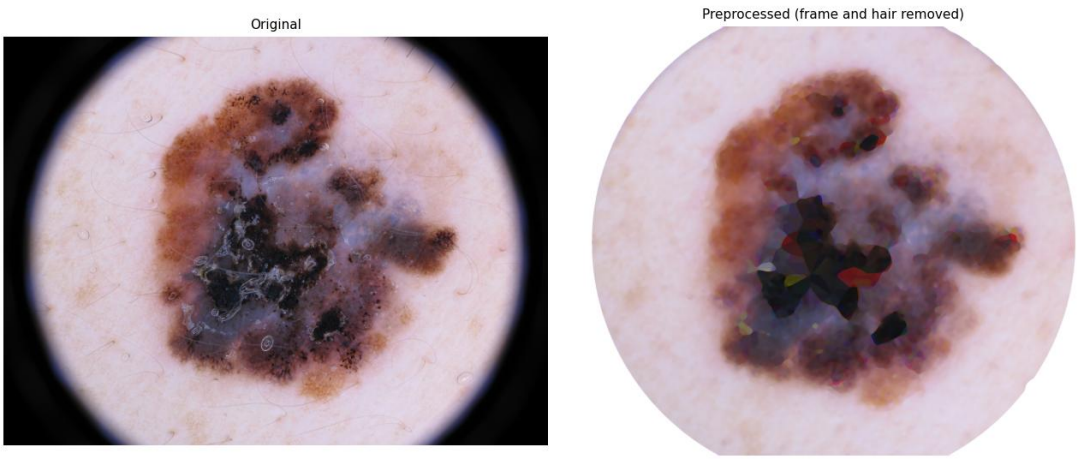


Figure 1: Preprocessing. Left: the raw dermoscopic image with its dark frame. Right: after frame removal and hair removal. The corners are whitened and the image cropped to the dermoscope disc.

Discussion. This step proved robust on the majority of standard images. Simple thresholding can nevertheless show its limits when the frame is partially reflective or the illumination strongly heterogeneous. Integrating shading-correction techniques [2], which model illumination variation by a quadratic function, would be a natural improvement. For this project the current method was sufficient.

One consequence of whitening the corners deserves emphasis, because it caused a real failure later in the pipeline: the whitened area is *synthetic*, not skin. Any method fitting a global colour model over the whole frame will let those pixels form a cluster of their own. Section 3.2 returns to this.

2.2 Hair removal

Dermoscopic images often contain hairs crossing the lesion or the surrounding skin. These artefacts mislead segmentation algorithms, which read them as contours or as internal lesion structures.

Our approach is based on the reference algorithm DullRazor [1], which detects thin dark structures by morphological closing.

2.2.1 First approach: naive implementation and its limits

We first tested a standard implementation using thin linear structuring elements (one pixel wide) and a high global threshold (250) on the difference image.

Analysis of the failure. This showed critical limits on images with thick or low-contrast hairs.

1. **Scale problem.** The structuring element being thinner than some hairs, the closing operated *inside* the hair without removing it, making detection impossible.
2. **Threshold problem.** A threshold of 250 demanded near-maximal contrast (black on white). The actual intensity difference between a brown hair and skin is often between 20 and 50, so the resulting binary mask was mostly empty and no interpolation could take place.

2.2.2 Second approach: a robust pipeline

The improved version works as follows:

1. **Multi-directional closing.** Rather than thin lines, we use structuring elements 10 pixels wide and 50 long, at 0°, 45°, 90° and 135°, and take the pixelwise maximum of the closings. A closing erases any dark structure its element cannot fit inside, so a hair thinner than 10 pixels is erased whatever its orientation.

2. **Detection and dilation.** The hair mask is the absolute difference between the original and the closed image, thresholded at 40 instead of 250, since the difference between a brown hair and skin is often only 20 to 50. The mask is then dilated by a 20×20 square to include the blurred edges of thick hairs.

3. **Interpolation and smoothing.** Masked areas are reconstructed by nearest-neighbour grid interpolation from the immediate neighbourhood, and a final median filter with a 20×20 window, applied to the whole image, blends the corrections in.

A coverage test guards the whole stage: the dilated hair mask is eroded by a disc of radius 12, so that isolated thin detections vanish, and if what survives covers less than 3 % of the valid area, the image is returned untouched. Inpainting a hairless image only blurs the very border the segmentation is looking for. An earlier version summed the raw intensity difference here instead of counting pixels, a quantity on a different scale from the fraction it was compared to; it now counts pixels.

2.2.3 What the detector actually detects

The detector responds to any dark structure narrower than some of its elements, and the dark, textured interior of a lesion is such a structure. Five of the twenty images pass the coverage test. Two are genuinely hairy (`ISIC_0000042` and `ISIC_0000095`) and one has a few thin hairs (`ISIC_0000146`). Of the other two, `ISIC_0000140` has only fine hairs and `ISIC_0000142` none, and on them almost all of the flagged pixels lie inside the lesion: the stage then inpaints parts of the lesion itself, which produces the small coloured shards visible in Figure 1.

The two diagonal elements were added on the belief that the axis-aligned ones missed diagonal hairs. They do not: every element is 10 pixels wide, so each of them already erases any thinner hair at any angle. What the diagonals add is more of the lesion's texture. A detector that separates hair from lesion texture, for instance by requiring elongation, is left as future work, and the scores reported here include this weakness.

3 Segmentation methods

Having cleaned the artefacts, the critical step is isolating the pigmented lesion from healthy skin. We implemented and compared three representative families of algorithms: a histogram-based method (Otsu), a texture-based method (LBP Clustering) and a region-merging method (SRM).

All three compute on a 256×256 copy of the image and upscale the resulting mask. The lesion border is a large-scale structure, so this loses no useful detail, and it keeps the three methods comparable by giving them the same amount of information.

3.1 Method 1: multi-channel Otsu thresholding

3.1.1 Theoretical principle

Otsu's method is the classical approach to image binarisation. It assumes the intensity histogram is bimodal (background and object) and searches for the threshold T minimising intra-class

variance, equivalently maximising the inter-class variance

$$\sigma_B^2(T) = \omega_0(T)\,\omega_1(T)\,\big[\mu_0(T) - \mu_1(T)\big]^2,$$

where ω_0, ω_1 are the class probabilities (skin/lesion) and μ_0, μ_1 their respective means.

3.1.2 Multi-channel combination

The performance of a global threshold depends drastically on the colour channel used [5]. A lesion may be invisible in the red channel, where skin and melanin reflect light similarly, yet strongly contrasted in blue.

Otsu's threshold is therefore computed independently on the red, green and blue channels, and the three binary masks combined as

$$M = (M_R \wedge M_G) \vee M_B.$$

A pixel is lesion when red *and* green agree, or when blue alone is confident. The asymmetry is deliberate: melanin absorbs short wavelengths preferentially, so blue carries the strongest melanin contrast and is trusted on its own, whereas red and green are required to corroborate each other.

The resulting mask has ragged borders. It is therefore used as the initial level set of a morphological Chan-Vese active contour, run for five iterations, which relaxes the border onto the actual intensity boundary.

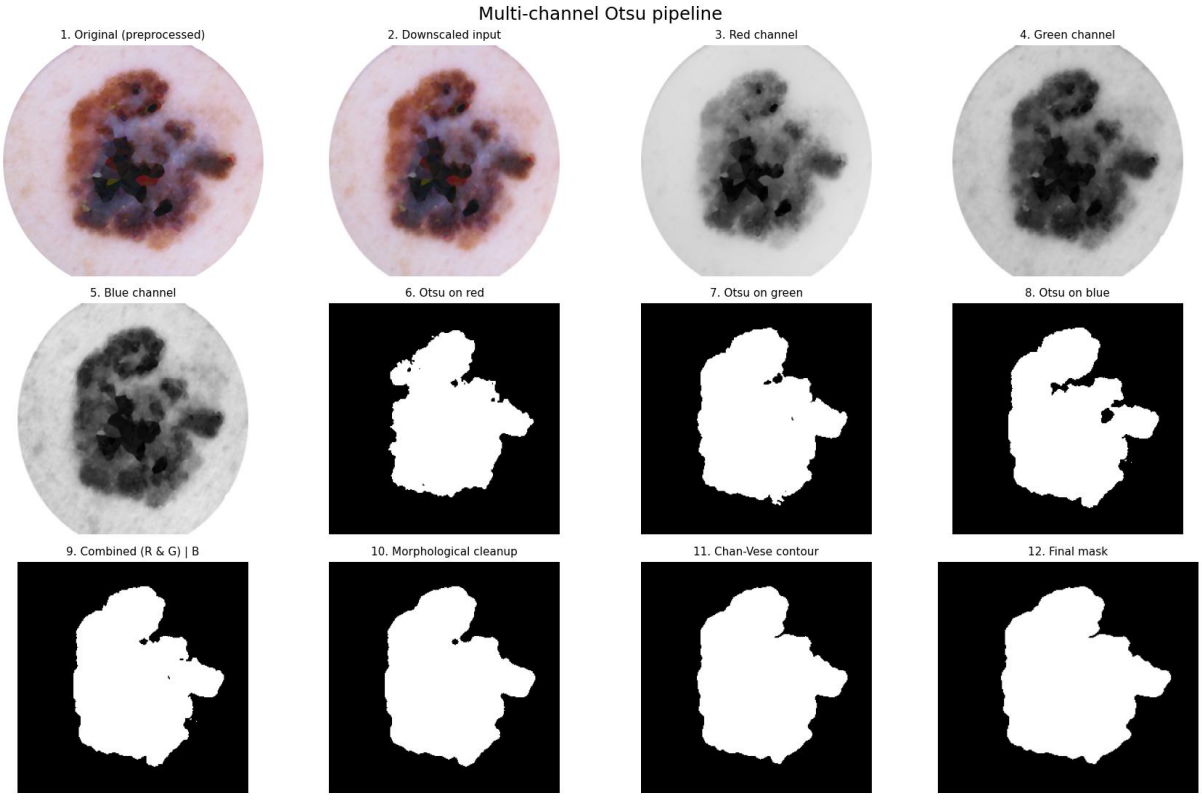


Figure 2: The multi-channel Otsu pipeline, stage by stage: the three colour channels, their individual Otsu masks, the combination, the morphological cleanup and the Chan-Vese refinement.

3.1.3 Observed limitations

Despite its speed, the method shows intrinsic limits on our dataset. Being global, it fails when the image presents strong vignetting or cast shadows, the shadow being segmented as part of the

lesion. For early or very pale melanomas the histogram is not clearly bimodal, and the computed threshold merges the lesion with the surrounding skin, causing critical under-segmentation.

3.2 Method 2: LBP Clustering

Simple thresholding fails when the lesion colour is too close to that of the skin. We therefore implemented a texture-based method following Pereira et al. [3], whose founding assumption is that healthy skin has a smooth, regular texture whereas a lesion has a chaotic one, with its pigment network, globules and dots.

3.2.1 Local Binary Patterns and pattern-subset binarisation

The luminance Y is computed according to ITU-R BT.601,

$$Y = 0.299\,R + 0.587\,G + 0.114\,B.$$

For each pixel, the Local Binary Pattern with $P = 8$ neighbours at radius $R = 1$ encodes the sign of the difference between each neighbour and the centre, packing the eight bits into a code in $[0, 255]$. The comparison is strict, matching the paper's $s(u) = \mathbf{1}[u > 0]$.

The key step is the binarisation on a *subset* of patterns. Code 0 (no neighbour brighter) and the eight powers of two (exactly one brighter neighbour) describe locally smooth neighbourhoods; every other code describes texture. Mapping the smooth subset to 0 and all others to 1 yields a texture map that is mostly ones, slightly denser inside the lesion (93 % vs 90 % of pixels).

3.2.2 Pseudo-RGB construction and the pinkness criterion

The binary texture map is smoothed by a Gaussian filter ($\sigma = 3$) and normalised into a continuous field $L \in [0, 1]$. The method then builds a *synthetic* image by stacking

$$\text{pseudo-RGB} = [\,L,\ Y,\ L\,],$$

and converts it to CIE L*a*b*, keeping only the chromaticity components (a^*, b^*). This is the elegant part of the method: a texture quantity and a luminance quantity are deliberately placed in a colour space so that a *colorimetric* criterion can separate them.

A k-means++ clustering with $k = 2$ partitions the pixels in that (a^*, b^*) plane. Which of the two clusters is the lesion is decided without supervision by the *pinkness* score

$$\mathcal{P} = \text{mean}\big[\max(a^*, 0) - \min(b^*, 0)\big],$$

the cluster with the higher score being retained. A morphological opening finishes the mask.

3.2.3 Interaction with frame removal

Applying the method as published to our cropped images failed completely on all three framed ones: the "lesion" it selected was the whole disc (raw Dice 0.62 to 0.76, and exactly zero with the inversion guard of Section 3.5.1, since removed). The cause is the interaction announced in Section 2.1: frame removal whitens the corners, and a white pixel has $L = 0$ texture with $Y = 255$, so in the pseudo-RGB it becomes *pure green*, an extreme point in the (a^*, b^*) plane. Those synthetic pixels captured one of the two clusters outright, and k-means separated disc from corners instead of lesion from skin.

Restricting the clustering to the valid disc area resolves it: the three images then score 0.708, 0.895 and 0.906. The lesson generalises beyond this project: a preprocessing step that fabricates pixel values must publish which pixels it fabricated.

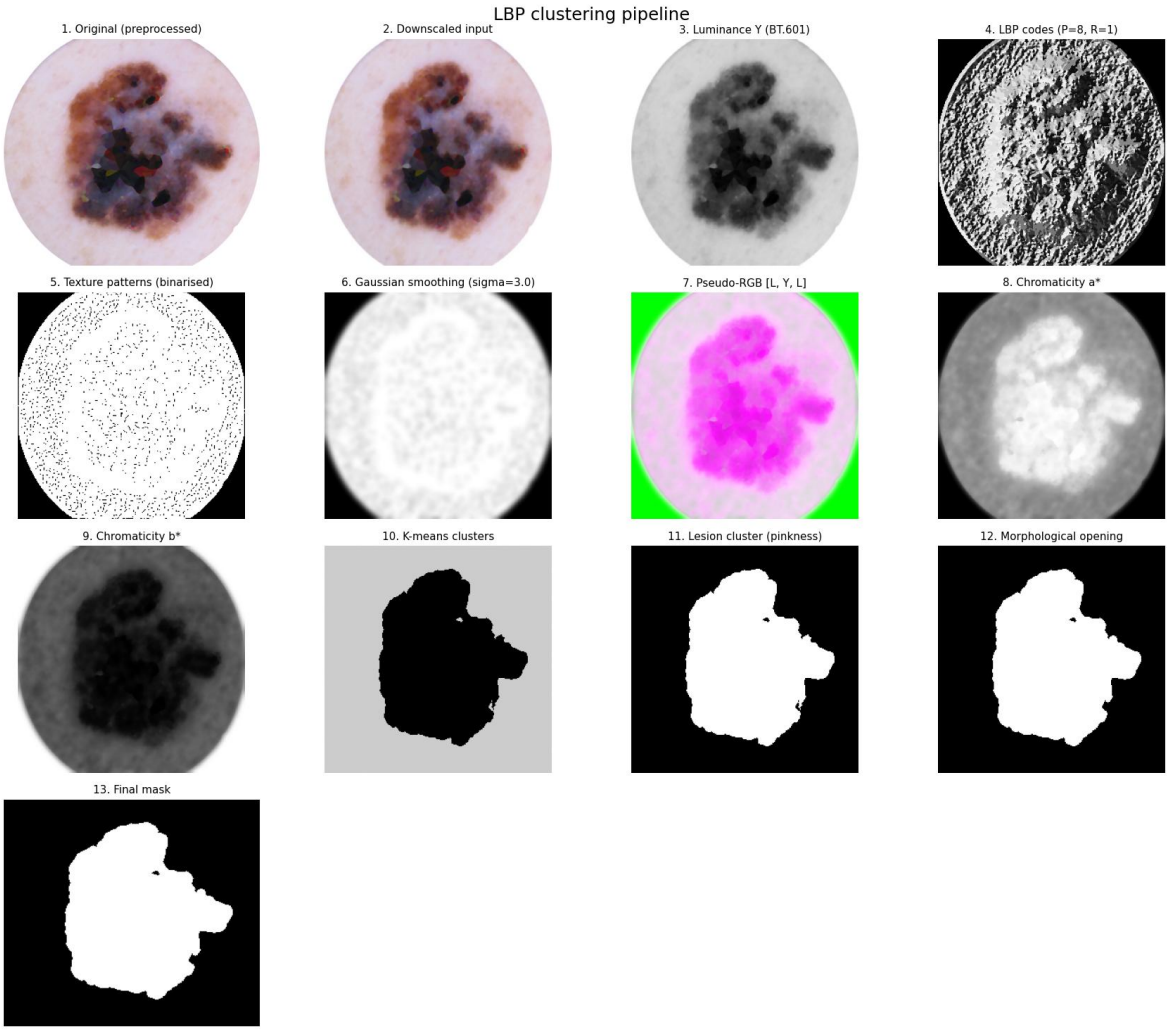


Figure 3: The LBP Clustering pipeline: luminance, LBP codes, the binarised texture patterns, the smoothed field L, the pseudo-RGB $[L, Y, L]$, its chromaticity components, the k-means clusters and the cluster selected by the pinkness criterion.

3.3 Method 3: Statistical Region Merging

SRM [8, 4] is a region-growing technique. Every pixel starts as its own region, and adjacent regions are merged iteratively when they are statistically similar.

3.3.1 The merging predicate

Pairs of 4-connected neighbouring pixels are sorted by increasing absolute intensity difference and visited in that order, so the most obviously similar neighbours merge first. Two regions R_1 and R_2 are merged when

$$\left(\overline{R_1} - \overline{R_2}\right)^2 < b(R_1) + b(R_2), \qquad b(R) = \frac{g^2}{2Q\,|R|}\Big(\ln(1 + |R|) + 2\ln(6n)\Big),$$

where $\overline{R}$ is the region mean, $|R|$ its size, n the pixel count and $g = 1$ for intensities in $[0, 1]$ (a simplification of Nock and Nielsen's bound). The bound tightens as regions grow: early merges are cheap, late merges must be well supported. The scale parameter Q controls permissiveness, a smaller Q yielding fewer, larger regions. Regions are maintained in a union-find structure with path compression and union by rank.

The implementation follows the algorithm directly rather than calling a library, and the smoothing is applied to the grayscale image, not to the colour image: a scalar Gaussian σ on an $(H, W, 3)$ array also blurs *across* the channel axis, mixing red into blue, and it measurably degraded the region scoring before it was found. The Felzenszwalb backend, kept for comparison, had the same mistake; it now smooths each colour channel separately.

3.3.2 From regions to a lesion

SRM produces an over-segmentation, not a lesion. Each region is scored on three criteria, combined as $0.5\,c + 0.3\,z + 0.4\,s$:

- **Contrast** c against skin. The skin reference is the 70th percentile of a one-pixel band along the image border, following the observation of Zortea et al. [6] that this band is mostly healthy skin. That holds on unframed images; on framed ones it does not (Section 3.4). The blue channel is used, for the melanin-absorption reason given in Section 3.1.
- **Centrality** z, a prior that the lesion lies near the centre.
- **Saturation** s, the HSV saturation above the skin median, which detects erythematous areas that are bright but pathological.

Polychrome lesions, a dark core with a brown periphery, are split by the over-segmentation, and keeping only the best region under-estimates the area. We therefore merge the winning region with every region scoring above 60 % of the maximum *and* genuinely darker or more saturated than skin. The second condition prevents the mask from creeping into shaded skin.

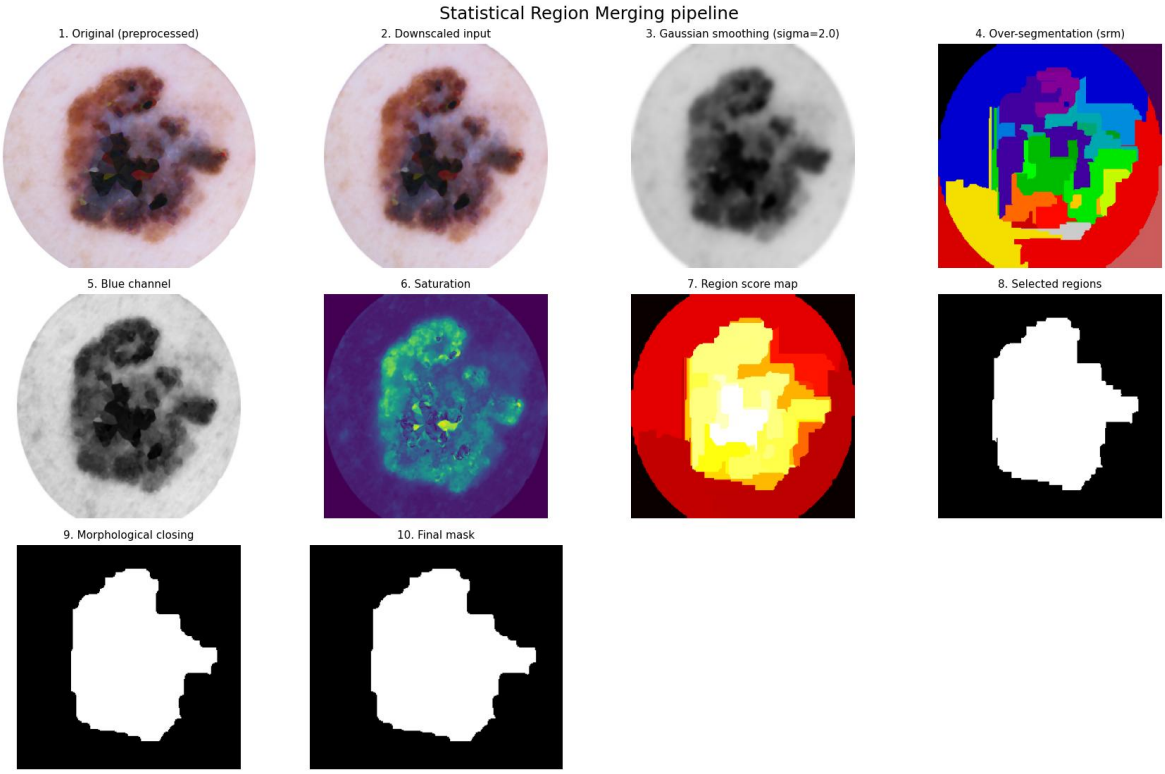


Figure 4: The SRM pipeline: smoothing, the union-find over-segmentation, the blue and saturation channels driving the scoring, the resulting score map, the selected regions and the final mask.

3.4 Otsu and SRM's blind spot to the frame corners

Section 3.2.3 covers why LBP's clustering has to ignore the whitened frame corners left by frame removal. Otsu and SRM have the same vulnerability. Otsu's per-channel threshold reads the corners as an artificially bright cluster. SRM reads its skin reference off a one-pixel band along the image border, and on a framed image, whose crop is the disc's bounding box, most of that band runs through the whitened corners (65–72 % of it on this dataset's three framed images), so the "skin" reference is largely the white of the corners.

Otsu's threshold is now computed on the valid pixels only. This does not raise Dice: on the three framed images it lowers it very slightly, by 0.001 to 0.006. It is kept because a threshold meant to separate lesion from skin should be computed on lesion and skin only.

The same change to SRM's reference lowers its Dice on all three framed images, by 0.07 to 0.21, because the hand-tuned weights and thresholds that judge a region's score were calibrated against the biased reference. Fixing it properly means retuning those weights, which is out of scope here. `region_merging.segment` keeps the corrected reference as an opt-in parameter, tested in isolation, and the benchmark does not enable it.

3.5 Post-processing: the smart convex hull

Using a convex hull to smooth borders is common practice but risky. On some images a small residual artefact in a corner produced an aberrant triangular mask spanning the whole image.

Before computing the hull, our `smart_convex_hull` filter analyses the connected components of the mask. It always keeps the largest, and keeps a satellite component only if it lies close to it, within a quarter of the image diagonal (Section 3.5.1 explains both choices). Distant specks of noise are discarded, so the convex shape stays meaningful.

The same post-processing (clipping to the valid area, hole filling and this hull) is applied identically to all three methods, so that the comparison isolates the segmentation itself.

3.5.1 Post-processing guards that were not doing what they claimed

An audit of this stage found its heuristics measuring the wrong quantity, silently, in four places. None of them raised an error. All four are fixed or removed in the version of the code this report describes.

`clean_mask` clipped to a guessed disc, not the real one. The dermoscope disc is known exactly: frame removal (Section 2.1) already computes it and returns it as a valid-area mask. `clean_mask` did not use it. Instead, on *every* image, it inscribed a circle in the array's own bounding box, regardless of whether that image had a frame at all. Seventeen of the twenty images in this dataset have none, so on those the guessed circle is fiction, and every lesion pixel that happened to fall outside it was deleted before scoring. On `ISIC_0000049`, whose lesion reaches toward a corner, applying the old guessed-disc crop to the ground truth itself erases 20.4 % of its area, a loss with nothing to do with segmentation quality, incurred before any method is even compared to the truth. The fix threads the real valid-area mask from preprocessing through to `clean_mask`, so the crop matches the disc that was actually removed, or clips nothing at all on an unframed image.

An inversion guard that could only do harm. `clean_mask` also flipped any mask covering more than 60 % of the valid area, on the theory that a segmenter covering that much had selected skin instead of lesion. On this dataset no segmenter output covers more than about 45 % of the valid area, while two real lesions (`ISIC_0000030` and `ISIC_0000049`) cover more than 60 %. The guard therefore never rescued anything, and would have turned a perfect segmentation of those two lesions into its complement. It was removed, and a unit test now checks that a large lesion comes through post-processing intact.

The hull's satellite reach grew with the lesion, past the image itself. A satellite blob was kept if it lay within 2.5 lesion radii of the main component, a reach that scales with the lesion. On a large lesion that reach can exceed the image: on `ISIC_0000145`, it worked out to 1376 px, while the single farthest point in the image (a corner) sits at only 1327 px. Every pixel therefore counted as "close enough", and the filter had nothing left to reject on that image, defeating the purpose stated at the top of this subsection. The fix measures the reach as a fraction of the image diagonal (0.25×) instead of the lesion's own radius. Across this dataset the farthest corner sits at 51–58 % of the diagonal regardless of lesion size, so this reach stays below that on every image while remaining generous enough to keep a genuinely nearby fragment.

A satellite kept for being large had no distance limit at all. The reach fixed above applied only to a satellite kept for being *close*; one kept for being *large* (more than 3 % of the main component's area) was exempt from any distance check, on the reasoning that a sizeable second lobe of the same lesion should count regardless of where it sits. That reasoning fails in practice: on `ISIC_0000095`, the LBP method's texture criterion also flags patches of scaly skin scattered across the image, far from the lesion, and the old rule glued the large ones into the hull. The buggy hull scores just 0.189 Dice on this image, and the fix (distance alone) recovers it to 0.694. Being large gave a region no more claim to belonging to the lesion than being small, so every satellite is now judged by the same reach, however large it is.

Why these went unnoticed. All four are silent by construction: a guard that never fires looks identical, in code and in a spot check, to a guard that never needed to fire. They surfaced only by asking, of each threshold, what quantity it was actually being compared against on this dataset's real image shapes and real segmenter outputs.

4 Results and discussion

4.1 Quantitative analysis

Each predicted mask is compared to the expert ground truth by the Dice coefficient. Table 1 gives the means over the 20 images.

Scoring a cropped image against its ground truth. Frame removal crops the three framed images to the disc's bounding box, so their masks are narrower than the ground truth. An earlier version of the pipeline resized the full-size ground truth to the cropped shape, squeezing it sideways instead of cutting the same columns off: prediction and truth no longer described the same pixels, and a *perfect* segmentation of those images scored only 0.915 to 0.965. The ground truth is now cropped with the exact box frame removal uses, and a unit test checks that a perfect mask on a framed image scores 1. The fix lowered every method's raw score on all three framed images, by up to 0.066 (the stretched truth happened to flatter them), and each method's mean raw Dice by 0.002 to 0.006. The ranking is unchanged. Every number in this report is measured after the fix.

Method	Dice (mask)	Dice (with convex hull)
Multi-channel Otsu	0.832	0.887
LBP Clustering	0.841	0.901
Statistical Region Merging	0.820	0.869

Table 1: Mean Dice coefficient per method.

Two trends stand out.

The convex hull helps all three methods by a similar margin. It raises the mean Dice of Otsu, LBP and SRM alike by roughly 0.049 to 0.060. It is not a uniform improvement: on three images (`ISIC_0000001`, `ISIC_0000145` and `ISIC_0000080`) it costs LBP between 0.012 and 0.027 Dice, because the raw mask was already close to the lesion's true shape and the added convexity is pure loss there. Convexity is an *assumption* about lesion shape, and it does not hold uniformly.

Texture gives the best raw masks, by a narrow margin. LBP Clustering produces the best masks before post-processing, ahead of Otsu and SRM, and with the hull applied it is also the best method overall. On twenty images these gaps are small, so the ranking is indicative rather than definitive.

4.2 Melanoma versus nevus

	Melanoma		Nevus	
Method	mask	+ hull	mask	+ hull
Multi-channel Otsu	0.815	0.872	0.849	0.902
LBP Clustering	0.842	0.895	0.841	0.908
Statistical Region Merging	0.790	0.849	0.850	0.888

Table 2: Mean Dice coefficient per method and per diagnosis.

Melanomas score lower than nevi on average in five of the six columns, which is the clinically expected direction: irregular, poorly defined borders are precisely what characterises melanoma. With ten images per class, however, none of these gaps is statistically significant (two-sided Mann–Whitney test, $p > 0.4$ in every column), and LBP's raw masks score the same on both diagnoses (0.842 against 0.841). This dataset cannot establish that melanomas are harder to segment.

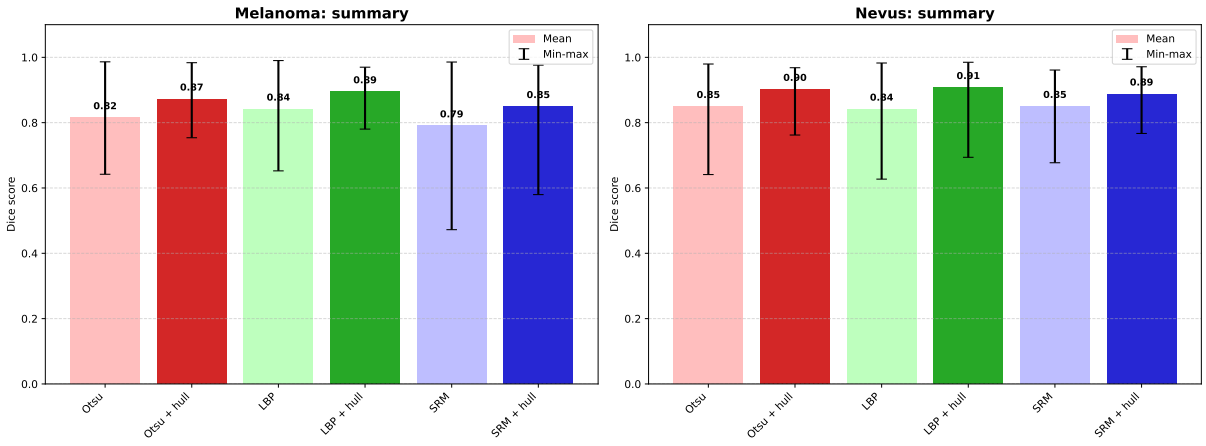


Figure 5: Mean Dice per method with min–max whiskers, melanoma (left) and nevus (right). SRM's spread is much wider on melanomas, down to 0.47 on `ISIC_0000049`; Otsu's is similar on both, and LBP's is wider on nevi, where `ISIC_0000095` pulls it down.

4.3 Failure analysis

The per-image scores in Table 3 make the failure modes explicit.

Two images drag every method's raw mask below 0.70. ISIC_0000024 is a low-contrast lesion where neither an intensity nor a texture cue separates lesion from skin cleanly. ISIC_0000049 is also the lesion covering the largest share of its image (66 %), and every method captures only part of it; whether its contrast or its size is to blame (the centrality prior and the border-sampled skin reference both assume a lesion well inside the image) is not established here. The convex hull recovers these two unevenly, lifting LBP above its own average on ISIC_0000024. These are the cases where a learned model, given enough annotated data, would most likely do better.

Image	Class	Otsu	+hull	LBP	+hull	SRM	+hull
ISIC_0000030	melanoma	0.66	0.75	0.71	0.81	0.70	0.82
ISIC_0000046	melanoma	0.87	0.93	0.78	0.89	0.72	0.78
ISIC_0000049	melanoma	0.64	0.75	0.65	0.78	0.47	0.58
ISIC_0000140	melanoma	0.88	0.93	0.90	0.94	0.89	0.95
ISIC_0000142	melanoma	0.90	0.92	0.92	0.94	0.93	0.95
ISIC_0000143	melanoma	0.71	0.77	0.77	0.83	0.70	0.75
ISIC_0000145	melanoma	0.99	0.98	0.99	0.97	0.99	0.98
ISIC_0000146	melanoma	0.88	0.94	0.90	0.96	0.83	0.90
ISIC_0000150	melanoma	0.81	0.88	0.88	0.90	0.81	0.88
ISIC_0000151	melanoma	0.83	0.87	0.92	0.93	0.88	0.91
ISIC_0000000	nevus	0.93	0.95	0.98	0.98	0.90	0.94
ISIC_0000001	nevus	0.98	0.96	0.88	0.86	0.95	0.93
ISIC_0000008	nevus	0.93	0.97	0.97	0.98	0.95	0.97
ISIC_0000019	nevus	0.87	0.91	0.90	0.93	0.85	0.89
ISIC_0000024	nevus	0.64	0.76	0.65	0.91	0.68	0.77
ISIC_0000042	nevus	0.80	0.85	0.84	0.90	0.72	0.79
ISIC_0000045	nevus	0.88	0.91	0.91	0.95	0.86	0.89
ISIC_0000080	nevus	0.94	0.96	0.97	0.96	0.96	0.96
ISIC_0000095	nevus	0.81	0.90	0.63	0.69	0.85	0.89
ISIC_0000112	nevus	0.71	0.85	0.68	0.91	0.79	0.84

Table 3: Dice coefficient per image and per method.

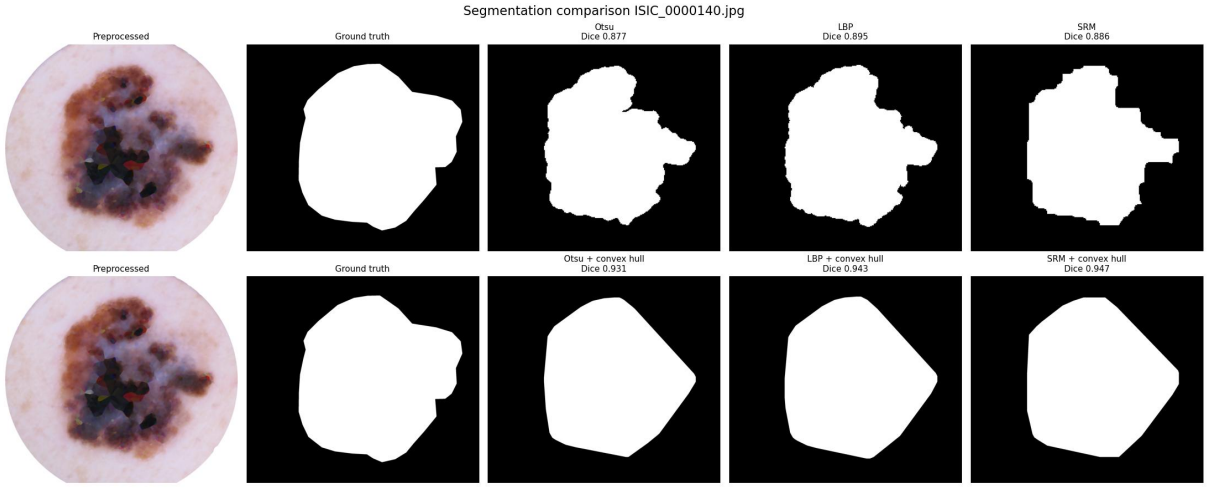


Figure 6: The three methods against the ground truth on one melanoma, with and without the convex hull.

4.4 Limitations

These numbers should be read with their limits in mind.

- **Twenty images is a demonstration, not an evaluation.** No confidence interval computed on this sample would be meaningful.
- **Parameters were tuned on these same images.** There is no held-out set, so the reported scores are optimistic.
- **The hair detector also detects lesion texture.** On two images (one with fine hairs only, one hairless) it inpaints parts of the lesion itself (Section 2.2.3), and the scores include that weakness.
- **A single ground truth per image.** Expert delineations vary between dermatologists, and the Dice ceiling attainable is bounded by that variability, which we cannot measure with one annotation.

5 Conclusion

We implemented a complete skin-lesion segmentation pipeline using classical computer vision only, from artefact removal through to quantitative evaluation, and compared three methods: two reimplemented from their source publications and a multi-channel Otsu baseline of our own design.

The outcome contradicts our initial expectation. We had assumed the most sophisticated method, region merging, would dominate; it does not. The texture-based LBP Clustering gives the best raw masks, and the simplest method, multi-channel Otsu, is highly competitive once the convex hull is applied. Algorithmic sophistication did not translate into accuracy on this dataset.

Two findings proved more instructive than the ranking itself. First, the interaction between frame removal and the LBP clustering, where a preprocessing step that fabricates pixel values silently destroyed a method that fits a global colour model, a class of bug invisible in the mean and obvious in the pipeline figures. Second, the cost of smoothing a colour image with a scalar Gaussian, which blurs across the channel axis and quietly corrupted the region scoring. Both were found by inspecting intermediate stages, which is exactly the affordance that a classical pipeline offers and a network does not.

The natural continuations are shading correction [2] to handle heterogeneous illumination, evaluation on a larger ISIC subset with a held-out parameter set, and using these masks as the input to the feature-extraction stage of a full CAD system.

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